V4.2 Toy Model 2C
Decoherence as Projection Stabilization

Rev A - Exploratory Quantum-Side Compatibility Report

Date: 2026.05.25 | *Continuation after Toy Model 2B*

Status: Toy Model 2C tests whether environmental record-coupling can suppress observer-accessible interference while preserving global unitary information. It is a compatibility model, not a solution to the measurement problem.

1. Purpose

Toy Model 2B showed that Bell correlations require a non-factorizable relational pair. Toy Model 2C now asks why the observer-accessible 3S+1T world appears classical even if the hidden relational state remains quantum-like.

The target is standard decoherence: interference terms become inaccessible to the local observer because the system becomes entangled with environmental records. In the 6D language, this is tested as projection stabilization.

2. Standard decoherence setup

Start with a two-state system:

```
|psi_S> = alpha |0> + beta |1>
```

The environment begins in an initial state:

```
|E_ready>
```

A measurement-like interaction correlates each system state with a different environment record:

```
(alpha |0> + beta |1>) |E_ready>
      -> alpha |0>|E_0> + beta |1>|E_1>
```

Term reminders:

• Environment means everything outside the small system that can carry records: air, photons, detector, surroundings.

• Record means a physical correlation that stores which alternative occurred.

• Unitary means the full system-plus-environment evolution preserves total information.

• Decoherence means interference becomes inaccessible in the reduced local description, even though the full state still exists.

3. The key quantity: environment overlap

The environment overlap is:

```
gamma = <E_1 | E_0>
```

Meaning: gamma measures how similar the two environment records are.

• If gamma = 1, the environment records are identical. No which-path information exists. Interference remains visible.

• If gamma = 0, the environment records are fully distinguishable. Which-path information exists. Interference disappears locally.

• If 0 < gamma < 1, decoherence is partial.

4. Reduced density matrix

The local observer does not track the full environment. The observer sees the reduced density matrix of the system:

```
rho_S = [[ |alpha|^2,        alpha beta* gamma ],
         [ alpha* beta gamma*, |beta|^2        ]]
```

The diagonal terms are the ordinary outcome probabilities. The off-diagonal terms are the visible interference terms.

As gamma approaches zero, the off-diagonal terms vanish:

```
rho_S -> [[ |alpha|^2, 0 ],
          [ 0, |beta|^2 ]]
```

Plain meaning: the observer sees a classical probability mixture, even though the full system plus environment state remains globally pure.

5. Projection-stabilization interpretation

Inside the 6D framework, Toy Model 2C interprets decoherence as follows:

```
hidden relational superposition
      -> coupling to environment record field
      -> environment alternatives become distinguishable
      -> gamma decreases
      -> observer-accessible interference terms vanish
      -> 1T projection stabilizes into classical-looking records
```

This does not say that the global state is destroyed. It says the 1T observer loses access to phase relations once they are dispersed into the environment record field.

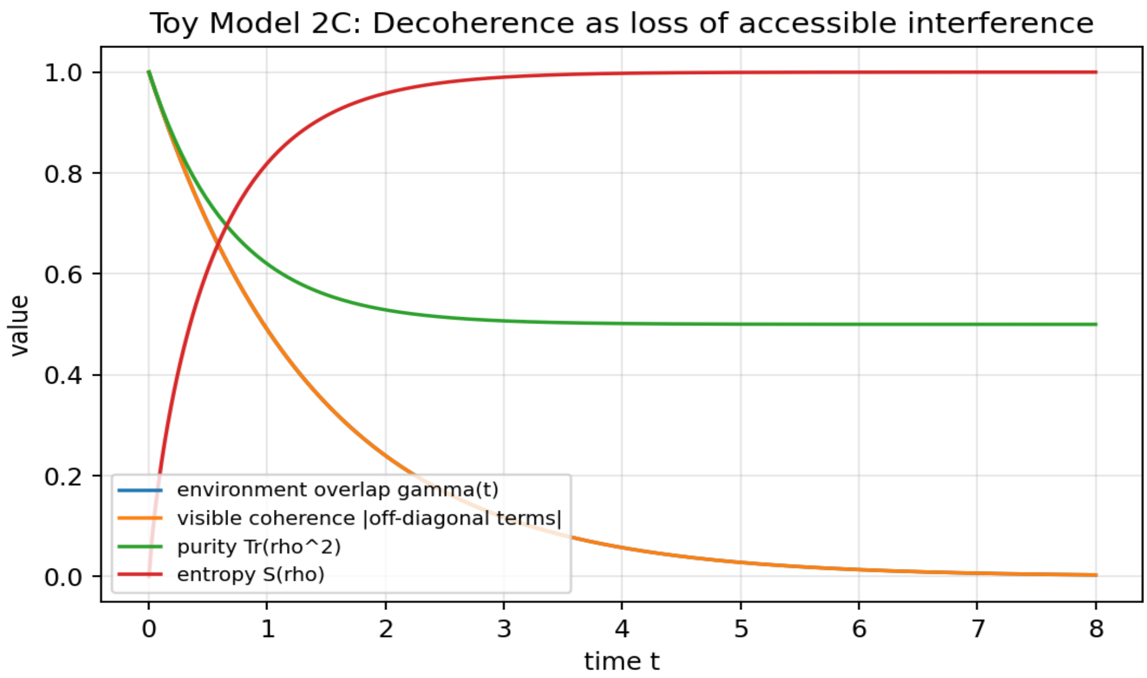
6. Numerical demonstration

For a balanced state alpha = beta = 1/sqrt(2), the simulation used:

```
gamma(t) = exp(-t / tau), with tau = 1.4
```

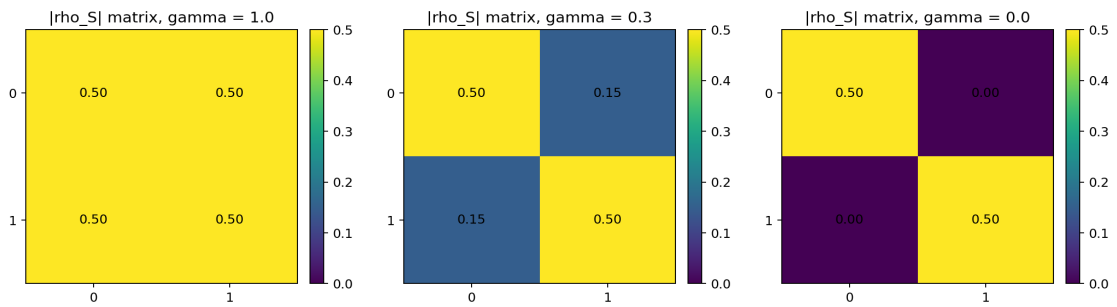
This is a simple decay model for environment-record distinguishability. It is not claimed as a fundamental law.

t	gamma	coherence	purity	entropy	P(0)	P(1)
0.0	1.000000	1.000000	1.000000	0.000000	0.500000	0.500000
0.5	0.699673	0.699673	0.744771	0.610250	0.500000	0.500000
1.4	0.367879	0.367879	0.567668	0.900046	0.500000	0.500000
2.8	0.135335	0.135335	0.509158	0.986747	0.500000	0.500000
5.6	0.018316	0.018316	0.500168	0.999758	0.500000	0.500000
8.0	0.003299	0.003299	0.500005	0.999992	0.500000	0.500000



7. Density matrix snapshots

These snapshots show the absolute values of the reduced density matrix. The off-diagonal entries shrink as gamma decreases.



8. Monte Carlo constraint checks

Random alpha, beta, and gamma values were sampled to test trace preservation, diagonal probability stability, Hermiticity, positivity, coherence scaling, and global normalization.

Check	Result
Random samples	10000
Max trace error	4.441e-16
Max diagonal probability error	0.000e+00
Max Hermiticity error	0.000e+00
Max positivity violation	0.000e+00
Max coherence-scaling error	1.110e-16
Max global norm error	4.441e-16
Max purity range violation	0.000e+00

9. What Toy Model 2C achieves

• It defines environment record coupling as the mechanism that suppresses visible interference.

• It shows mathematically that off-diagonal terms are multiplied by gamma = .

• It preserves the diagonal Born probabilities P(0) = |alpha|^2 and P(1) = |beta|^2.

• It preserves global unitarity at the system-plus-environment level.

• It produces classical-looking local records when gamma approaches zero.

• It passes trace, positivity, Hermiticity, coherence-scaling, and normalization checks.

10. What Toy Model 2C does not achieve

• It does not solve the measurement problem.

• It does not select a single actual outcome by itself.

• It does not derive the Born rule.

• It does not prove physical 3T existence.

• It does not produce a new experimental prediction.

• It does not yet connect decoherence to black holes, horizons, or dark-sector accessibility.

11. Important distinction: decoherence is not collapse

Decoherence explains why interference becomes locally invisible. It does not, by itself, explain why one specific outcome is experienced rather than another.

Therefore the correct statement for the framework is:

```
Decoherence-as-projection-stabilization can explain the emergence of classical-looking records,
but it does not yet explain single-outcome selection.
```

This guardrail is important. Without it, the framework would overclaim.

12. Close-out statement for 2C

Toy Model 2C successfully formalizes decoherence as projection stabilization. A hidden relational superposition coupled to an environment record field produces a reduced observer-accessible state whose off-diagonal interference terms are suppressed by the environment overlap gamma. The model preserves global unitarity and local probability normalization while generating classical-looking local records. It does not solve measurement collapse or derive the Born rule. It is a successful compatibility model for decoherence within the reciprocal projection framework.

13. Recommended next chapter

The next logical chapter is Toy Model 2D: Measurement Update and Single-Outcome Constraint.

1. Define the difference between decoherence and actual outcome selection.

2. Test whether projection stabilization can be extended to an observer-record update rule.

3. Check whether any update rule violates unitarity or no-signalling.

4. Separate interpretation from mathematics: many-worlds, relational, collapse-like, or epistemic update.

5. Reject any version that allows controllable faster-than-light signalling or probability non-conservation.

14. Rebind prompt for continuation

```
We are resuming after Toy Model 2C.

2C tested decoherence as projection stabilization.

Standard setup:
(alpha|0> + beta|1>) |E_ready>
      -> alpha|0>|E_0> + beta|1>|E_1>

Environment overlap:
gamma = <E_1|E_0>

Reduced density matrix:
rho_S = [[ |alpha|^2,        alpha beta* gamma ],
         [ alpha* beta gamma*, |beta|^2        ]]

Result:
As gamma -> 0, off-diagonal interference terms vanish locally while diagonal
Born probabilities remain unchanged.

Interpretation:
hidden relational superposition
      -> environment record coupling
      -> phase information dispersed into record field
      -> 1T observer sees stabilized classical-looking records

Guardrail:
Decoherence is not collapse. 2C does not explain single-outcome selection,
does not derive the Born rule, and does not prove physical 3T existence.

Next recommended chapter:
Toy Model 2D - Measurement Update and Single-Outcome Constraint.
```